# Southeast University, Bangladesh

## CSE261: Numerical Methods

## Group Assignment Report

**Assignment Topic:** Multistep Methods (Adams–Bashforth Adams–Moulton)


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**Abstract**

This report presents the implementation and analysis of Multistep Methods (Adams–Bashforth  Adams–Moulton). It presents a numerical comparison of multistep methods for solving first-order ODEs, including Adams–Bashforth (AB2, AB3) and Adams–Moulton (AM2) methods, with a predictor–corrector approach for the implicit scheme. The fourth-order Runge–Kutta (RK4) method is used as a reference due to its strong accuracy and stability. All methods are tested on a problem with a known analytical solution, and error and solution plots are analyzed using a large step size to illustrate stability behavior. The results show the superior stability of RK4 and the conditional stability of multistep methods.

# 1 Introduction

Numerical methods for ordinary differential equations are essential when analytical solutions are difficult or impossible to obtain. Among these methods, single-step methods such as Runge–Kutta compute the solution using information from only the current point, whereas multistep methods utilize solution values from several previous points, improving computational efficiency. The Adams–Bashforth methods are explicit multistep schemes, while Adams–Moulton methods are implicit and generally more stable. However, multistep methods are known to suffer from stability issues, particularly for large step sizes.

# Objectives

The objectives of this assignment are:

1. To implement AB2, AB3, and AM2 (predictor–corrector) methods
2. To compare their performance with RK4 using identical step sizes
3. To visualize stability behavior and numerical errors
4. To analyze divergence and instability in multistep methods

# 2 Theoretical Background

Consider the first-order ordinary differential equation:

$$\frac{dy}{dt} = f(t, y), \quad y(0) = y_0$$

## Test Problem

$$\frac{dy}{dt} = -2y + e^{-t}$$

The exact solution is:

$$y(t) = e^{-2t} + \frac{1}{2}e^{-t}$$

This equation is well-suited for stability analysis because it is stiff-like and exhibits exponential decay.

### Runge–Kutta 4th Order (RK4)

RK4 is a single-step, fourth-order accurate numerical method given by:

$$y_{n+1} = y_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

RK4 is conditionally stable but significantly more robust than explicit multistep methods.

### Adams–Bashforth 3-Step (AB3)

$$y_{n+1} = y_n + \frac{h}{12}\left(23f_n - 16f_{n-1} + 5f_{n-2}\right)$$

- Explicit method
- Third-order accurate
- More sensitive to step size

### Adams–Moulton 2-Step (AM2) Predictor–Corrector

**Predictor (AB2):**

$$y_{n+1}^{(p)} = y_n + \frac{h}{2}\left(3f_n - f_{n-1}\right)$$

**Corrector (AM2):**

$$y_{n+1} = y_n + \frac{h}{2}\left[f\left(t_{n+1},\, y_{n+1}^{(p)}\right) + f_n\right]$$

- Implicit method
- Improved stability compared to Adams–Bashforth methods

## 3 Methodology

# Algorithm Steps

1. Define the ODE function and its exact solution
2. Generate starting values using RK4 (required for multistep methods)
3. Apply AB2, AB3, and AM2 predictor–corrector methods
4. Compute numerical solutions up to $t = 5$
5. Calculate absolute errors:

$$\text{Error} = |y_{\text{numerical}} - y_{\text{exact}}|$$

6. Plot:
   - Numerical solutions versus the exact solution
   - Error curves for all methods

## Why These Methods?

- Multistep methods are computationally efficient
- RK4 serves as a stable reference method
- A large step size ($h = 1.0$) is used to clearly demonstrate stability issues

# 4 Implementation

## Programming Tools and Code Structure

**Programming Language:** C++

**Tools Used:**

- STL vectors for data storage
- File I/O for CSV output

**Output:**

- CSV file (`multistep_results.csv`) for plotting in Excel

### Code Structure (Overview)

- `f(t,y)` – Defines the ODE
- `exact(t)` – Computes the exact analytical solution
- `RK4_step()` – Performs a single RK4 iteration
- `AB2()`, `AB3()` – Explicit multistep solvers
- `AM2_PC()` – Predictor–corrector implementation
- Error computation and CSV export

## Project Repository and Resources

The complete implementation and results for this project are available on GitHub:

- **GitHub Repository:** `https://github.com/Ornab10/TeamPuppis3.0`
- **Basic Project Information:** See the `README.md` file in the repository.
- **Source Code (C++):** `https://github.com/Ornab10/TeamPuppis3.0/blob/main/code3.0.cpp`
- **Graphs and CSV Data:** `multistep_results.csv`, `multistep_results.xlsx`

# 5 Results and Analysis

## Solution Behavior (Solution Graph)

From the solution graph:

- RK4 closely follows the exact solution throughout the interval
- AB2 shows increasing deviation after $t = 3$
- AB3 becomes unstable earlier, with oscillations and sign changes
- AM2 exhibits the most severe instability for large step sizes, diverging rapidly

This confirms that explicit multistep methods are highly sensitive to step size.

## Error Analysis (Error Graph)

From the error graph:

- RK4 maintains a low and smoothly increasing error
- AB2 shows moderate error growth
- AB3 error increases rapidly after $t = 3$
- AM2 exhibits the largest error, indicating predictor–corrector instability when the predictor is already inaccurate

At $t = 5$, the absolute errors are summarized as:

| **Method** | **Error** |
|---|---|
| RK4 | 0.00805 |
| AB2 | 0.06683 |
| AB3 | 0.74194 |
| AM2 | 0.76896 |

## Stability Discussion

- RK4 remains stable even for large step sizes
- AB2 is conditionally stable but deteriorates over time
- AB3 is more accurate in theory but less stable in practice
- AM2 stability depends strongly on predictor quality

These results clearly demonstrate the stability limitations of multistep methods.

# 6 Discussion

The results clearly indicate that:

- Multistep methods are highly sensitive to the choice of step size
- Explicit Adams–Bashforth methods suffer from instability for large $h$
- The implicit AM2 method, even with a predictor–corrector approach, may still diverge when the step size is too large
- RK4 remains robust and stable under identical conditions

## Strengths

- Efficient and well-structured implementation
- Clear demonstration of numerical instability
- Proper use of RK4 as a reference method

## Limitations

- Stability heavily depends on the choice of step size
- Multistep methods require accurate starting values

# 7 Conclusion

- RK4 provides the most stable and accurate results for large step sizes
- Adams–Bashforth methods exhibit instability when the step size $h$ is large
- The predictor–corrector AM2 method improves accuracy but does not fully eliminate instability
- Error and solution plots clearly visualize numerical behavior
- Reducing the step size significantly improves the performance of multistep methods